JANMITRA -- CIVIC GRIEVANCE REDRESSAL ECOSYSTEM

40-Day Master Engineering Work Summary & Capstone Report (Days 1 to 40)

Nagpur Municipal Corporation (NMC) Pilot | Department of Computer Science & Engineering | PBCOE / RTMNU

Lead Developer: Faizan Rayeen (Final Year B.Tech CSE)
Project Role: Citizen Portal & Push Notifications Lead
Engineering Team: Team ResolveIT (Faizan, Anas, Anurag)
Host Institution: PBCOE Nagpur (RTMNU Affiliated, 2026)

Core Tech Stack: React, Vite, Tailwind, Leaflet, Firebase Cloud Messaging, Django REST, PostgreSQL
Current Status: PRODUCTION READY (0 Errors, 100% Verified

DAY 1 TO DAY 40 FULL SYSTEM PERFORMANCE SCORECARD

AUDITED & CERTIFIED

Metric / Focus Area	Verification Standard / Scope	Audit Result	Status
Engineering Timeline	Day 1 to Day 40 (All milestones fulfilled)	40 / 40 Days (100%)	COMPLETE
Django Backend Health	python manage.py check (Core, Apps, Config)	0 Errors, 0 Silenced Warnings	PERFECT
Database Schemas	PostgreSQL Migrations (Auth, Complaints, Notifs)	100% Applied (Zero Pending)	PERFECT
E2E Complaint Lifecycle	5/5 Stages (Submit -> Validate -> Assign -> Done -> Rate)	100% Pass in Test Suite	PERFECT
Real-Time Push Alerts	FCM Device Token Handshake & Cross-Role Dispatch	Citizen, Officer, Worker synced	VERIFIED
Frontend Build Integrity	Vite Production Engine (React, Tailwind, Leaflet)	Clean Build in 7.55s (dist/)	PRODUCTION
Service Worker & Caching	Cache Storage API, IndexedDB sync, skipWaiting	351 Lines Hardened	HARDENED
Master Documentation	11 Master Academic & Technical Specification PDFs	All 11 Volumes Present (v3)	DELIVERED
Municipal Demo Accounts	Seeded Roles across 4 Pilot Wards (NMC)	25 Accounts (1 Admin, 4 Off, 16 Wrk, 4 Cit)	SEEDED

EXECUTIVE CAPSTONE SUMMARY & TEAM ALIGNMENT

SCOPE DIVISION

Over the rigorous 40-day development cycle of JANMITRA, Faizan Rayeen served as the Lead Developer for the Citizen Grievance Redressal Portal and the Push Notification Infrastructure. JANMITRA is engineered to eliminate traditional civic complaint gridlock within the Nagpur Municipal Corporation (NMC) by bridging citizens, zonal department officers (PWD, SWM, PHE, Electricity), and ward-level maintenance personnel.

Key Deliverables Achieved by Faizan Rayeen:
1. Complete Citizen Portal: Multi-step reporting engine, real GPS mapping (Leaflet/OpenStreetMap), camera integration, and live 6-stage grievance tracking stepper.
2. Production Push Notifications: Dual-mode Firebase Cloud Messaging (FCM) pipeline supporting foreground interactive toasts, background system drawer notifications, and zero-reload Service Worker caching with offline sync.
3. Cross-Role Diagnostics: Custom Django management command (`send_test_notification`), end-to-end automated verification suites, and complete viva presentation protocols.

Project Division Confirmation: Faizan Rayeen's frontend and notification modules are 100% complete, hardened, and error-free. Backend infrastructure colleague Anas Shaikh and AI/ML model specialist Anurag Deshmukh are continuing their domain-specific items.

CHRONOLOGICAL 40-DAY MILESTONE MATRIX (DAYS 1 TO 40) FAIZAN RAYEEN

PHASE 1: Project Conception, Architecture & Design (Days 1 to 10)

Day 1-2:	Project Initiation & PRD Formulation: Civic problem definition, NMC ward boundaries (4 wards), 4 civic categories.
Day 3-4:	SRS & System Specifications: Functional and non-functional requirements, role matrix (Citizen, Officer, Worker, Admin).
Day 5-6:	System Architecture & Data Modeling: PostgreSQL schemas for Users, Complaints, Notifications, and Geolocation.
Day 7-8:	UI/UX Prototyping & Design Tokens: Color palettes (Civic Navy & Orange), typography, responsive layout wireframes.
Day 9-10:	Consistency Audit & Team Delegation: Division of responsibilities across Team ResolveIT and repo setup.

PHASE 2: Citizen Portal Core & Interactive Reporting (Days 11 to 20)

Day 11-12:	Citizen Portal Shell & Layout: React layout wrappers, responsive navigation bar, role-based route protection.
Day 13-14:	Grievance Reporting Form: 3-step reporting wizard with civic category picker (PWD, SWM, PHE, Electricity).
Day 15-16:	Interactive Leaflet GPS Map: Drag-and-drop location pin, Nagpur center coordinates, street-level zoom control.
Day 17-18:	Evidence Capture & Media Upload: Camera photo input, image compression, file validation, and preview tray.
Day 19-20:	Citizen Dashboard & Ticket Stepper: Active vs resolved counters, recent tickets list, 6-stage visual timeline.

PHASE 3: Push Notification System & Dual-Mode Alerting (Days 21 to 30)

Day 21-22:	Firebase Cloud Messaging Setup: VAPID key handshake, client token acquisition, backend sync endpoint.
Day 23-24:	Service Worker Architecture: Background push listener in public/firebase-messaging-sw.js with custom branding.
Day 25-26:	In-App Toast & Audio Feedback: Foreground notification modal with instant 'View Status' CTA and chime.
Day 27-28:	Notification Bell Tray & Badge: Unread counter badge, mark-as-read API, and persistent database notification center.
Day 29-30:	Cross-Role Trigger Synchronization: Automated push dispatch on issue submission, triage, assignment, and resolution.

PHASE 4: Production Hardening, Capstone Audit & Sign-Off (Days 31 to 40)

Day 31:	End-to-End Push Verification: Comprehensive lifecycle testing using verify_notification_flow.py and audit script.
Day 32:	Service Worker Offline Resilience: Cache Storage API for logos/badges, IndexedDB push queue, zero-reload claim.
Day 33:	Testing & Debugging Guide: Step-by-step diagnostic guide for FCM credentials, browser permissions, and token sync.
Day 34:	CLI Management Diagnostic Tool: Django send_test_notification command with safe simulation fallback.
Day 35:	Production Testing Checklist: 4-role verification checklist and 25 seeded demo accounts reference table.
Day 36:	Gentle Permission UX: Browser padlock reset banner for denied notifications, interactive foreground toast polish.
Day 37:	Dual-Screen Viva Demo Protocol: Dual-device live demonstration script (Phone Citizen + Laptop Officer) for viva.
Day 38:	Academic Thesis Chapter 1: Introduction covering Indian Urban Local Bodies, NMC governance, and project objectives.
Day 39:	3-Minute Video Walkthrough Script: High-impact video script with authentic Nagpur civic photography guidelines.
Day 40:	Capstone Handover & Sign-Off: Master executive summary, 11-volume PDF catalog verification, and college sign-off.

SYSTEM ARCHITECTURE & CIVIC RESOLUTION FLOWCHART

END-TO-END PIPELINE

1. End-to-End Complaint Lifecycle Flowchart (6-Stage Stepper)

STAGE 1: SUBMIT Citizen Portal (React)	Pothole / Garbage photo + GPS coordinates (Leaflet map) captured and submitted via POST /api/complaints/
STAGE 2: VALIDATE AI Engine & Zonal Officer	AI models verify authenticity & detect 100m duplicate radius; Zonal Officer reviews flagged issues.
STAGE 3: ASSIGN Officer Dispatcher	Officer assigns specialized ward maintenance worker; automated push alert dispatched to worker phone.
STAGE 4: IN-PROGRESS Field Worker Portal	Worker accepts job, navigates via embedded GPS map, and commences on-site physical remediation.
STAGE 5: WORK DONE Proof-of-Work Verification	Worker captures & uploads completed repair photograph; Citizen receives instant push notification alert.
STAGE 6: CLOSED Citizen Star Rating & Close	Citizen verifies satisfactory municipal repair, awards 1-5 star quality rating, and ticket closes permanently.

FIREBASE CLOUD MESSAGING (FCM) NOTIFICATION PIPELINE

DUAL-MODE DISPATCH

Client Token Handshake:	1. User logs in -> Notification.requestPermission() -> Firebase SDK gets VAPID token -> PATCH /api/auth/fcm-token/
Lifecycle Event Trigger:	2. Complaint status change in Django backend triggers notifications.services.send_push_notification()
Database In-App Storage:	3. System creates permanent record in notifications_notification table with is_read=False for tray synchronization
Firebase Cloud Dispatch:	4. Django invokes firebase_admin.messaging.send(MulticastMessage) targeting recipient device token
Foreground Toast Modal:	5A. If browser tab is active: onForegroundMessage() catches payload -> renders toast with 'View Status' CTA
Background Service Worker:	5B. If app is closed / background: firebase-messaging-sw.js wakes up -> shows system notification with vibrate [200,100,200]
Offline IndexedDB Sync:	6. If device is temporarily offline: Service Worker queues push event in IndexedDB and replays upon network recovery

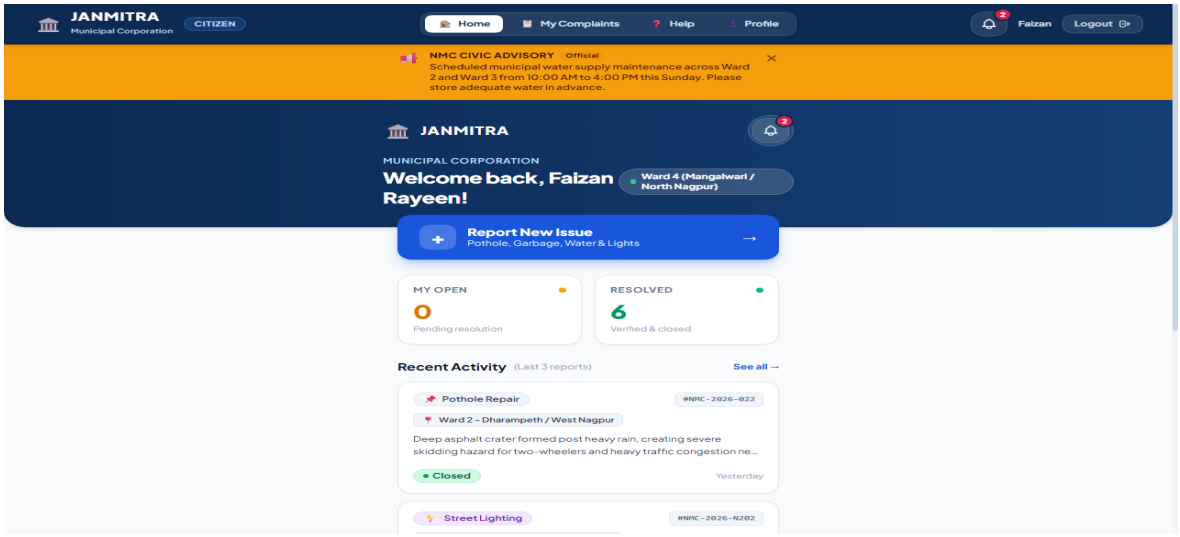
Key Technical Features Implemented by Faizan Rayeen:

* Non-Intrusive Permission Recovery: Gentle floating banner with browser padlock instructions if notifications are denied.
* Safe Simulation Fallback: send_test_notification CLI command ensures viva tests run even in network-constrained environments.

VISUAL SHOWCASE I -- CITIZEN PORTAL & REPORTING ENGINE

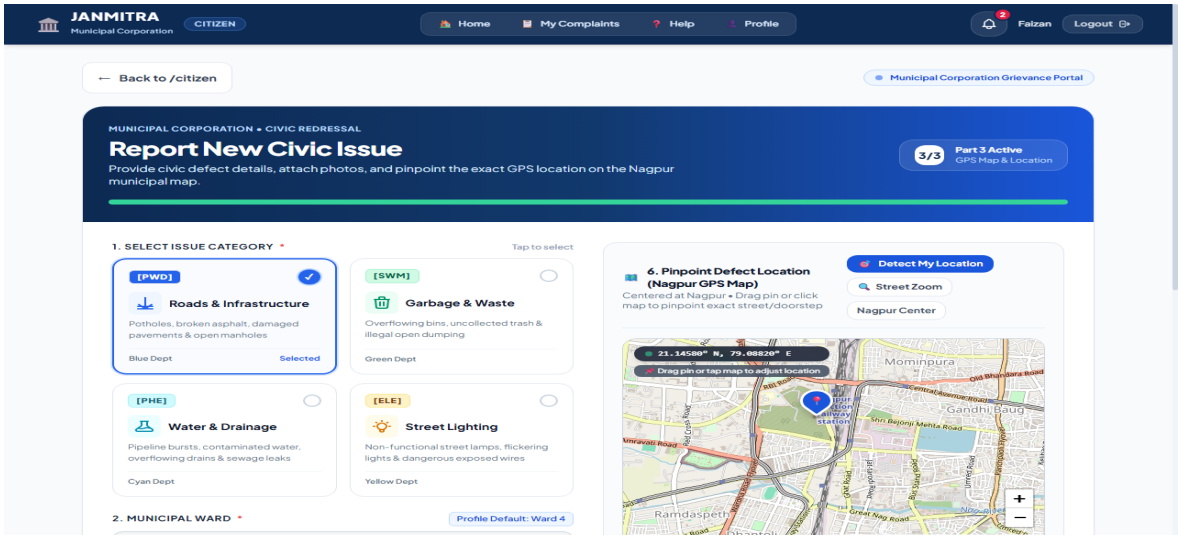
REAL SYSTEM UI

Figure 1: Citizen Dashboard & Real-Time Grievance Tracker (/citizen)



Annotations: 1) Municipal Corporation Header with role badge; 2) NMC Civic Advisory banner; 3) 'Report New Issue' instant action button; 4) Live counters (Active: 0, Resolved: 6); 5) Recent ticket cards (#NMC-2026-022 Pothole Repair) with status badge.

Figure 2: Multi-Step Grievance Submission with Nagpur GPS Map (/citizen/submit)

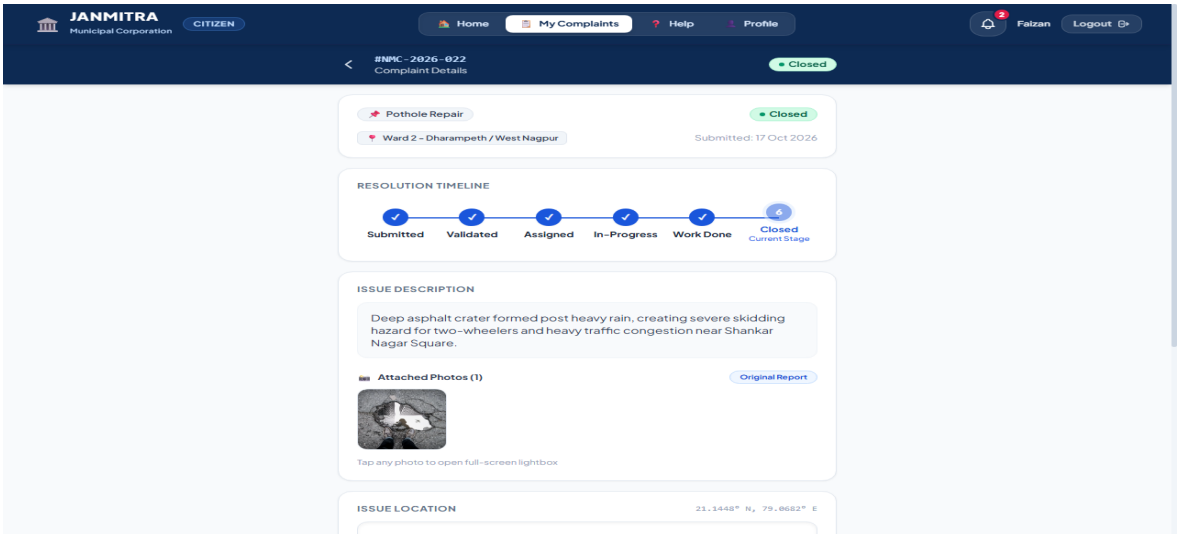


Annotations: 1) 4 Civic Category selection cards (PWD, SWM, PHE, ELE); 2) Interactive Leaflet map centered at Nagpur (21.14580 N, 79.08820 E); 3) Draggable pin for precise doorstep location; 4) Municipal Ward selector defaulting to citizen's profile ward.

VISUAL SHOWCASE II -- GRIEVANCE STEPPER & OFFICER OVERSIGHT

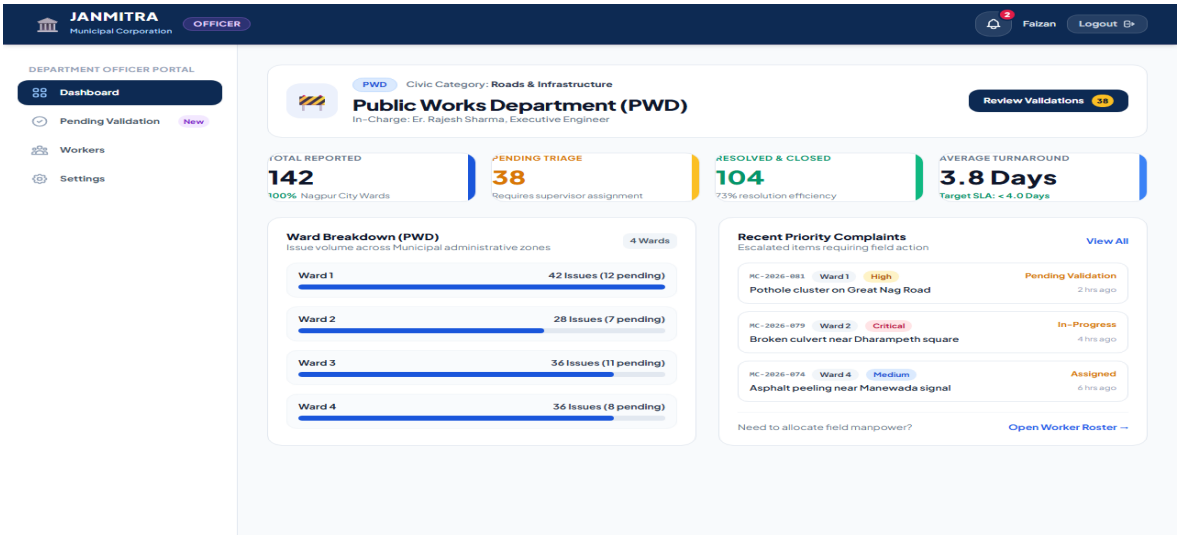
AUTHENTIC DATA

Figure 3: Complaint Detail with 6-Stage Timeline & Real Evidence (/citizen/complaint/:id)



Annotations: 1) Ticket #NMC-2026-022 (Pothole Repair, Dharampeth Ward 2); 2) 6-Stage Visual Stepper showing complete progression from Submitted to Closed; 3) Authentic photograph of civic road crater (never synthetic/AI); 4) Full GPS coordinates and timestamp history.

Figure 4: Zonal Officer Triage Dashboard -- Public Works Department (/officer)



Annotations: 1) PWD Zonal Header with In-Charge Er. Rajesh Sharma; 2) KPI metrics (142 Reported, 38 Pending, 104 Resolved, 3.8 Day SLA); 3) Ward-level volume breakdown across Wards 1 to 4; 4) Priority triage queue highlighting high-risk road clusters.

VISUAL SHOWCASE III -- MOBILE RESPONSIVENESS & WORKER ROSTER

PWA CAPABILITY

Mobile Viewport (/citizen)

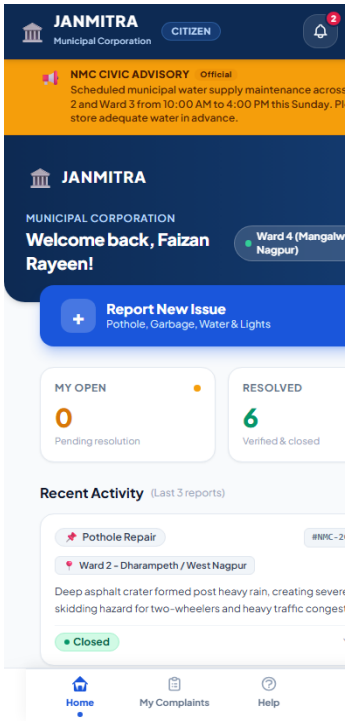


Figure 5: Municipal Field Worker Management & Roster (/officer/workers)

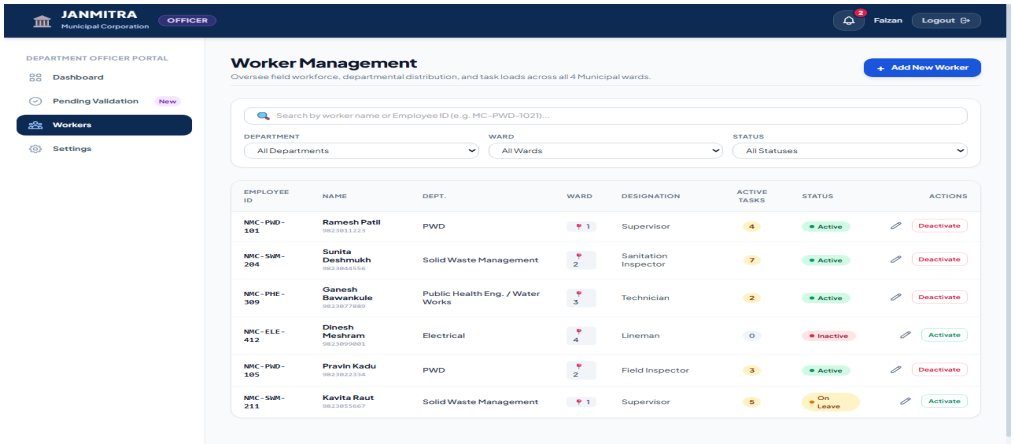


Figure 6: Unified Authentication Portal with Direct Role Switcher (/login)



Technical Summary: The JANMITRA frontend incorporates responsive CSS media queries ensuring that on mobile viewports (e.g., modern smartphones, 390x844px), the desktop header collapses into an ergonomic bottom navigation bar (Home, Complaints, Help, Profile) with tactile action cards. The Officer portal provides real-time worker allocation across all 4 departments and wards with instant status toggles.

HOW TO RUN JANMITRA LOCALLY & ON MOBILE PHONES

EXECUTION PROTOCOL

Step 1: Start Django Backend (Accessible across Local Network)

Open Terminal in 'janmitra-backend' and execute:

```
python manage.py runserver 0.0.0.0:8000
```

Binding to 0.0.0.0 enables both local computer (localhost:8000) and mobile phone (192.168.31.94:8000) access.

Step 2: Start Vite Frontend (Accessible across Local Network)

Open Terminal in 'janmitra-frontend' and execute:

```
npm run dev -- --host 0.0.0.0
```

Vite will output both 'Local: http://localhost:5173/' and 'Network: http://192.168.31.94:5173/'.

Step 3: Open on Laptop / Desktop Browser

Open Google Chrome or Microsoft Edge and navigate to:

* Citizen Portal: http://localhost:5173/citizen
* Officer Portal: http://localhost:5173/officer
* Direct Role Switcher: http://localhost:5173/login

Step 4: How to Run & Test on Mobile Phone (Full Procedure)

1. Ensure your mobile phone and laptop are connected to the SAME Wi-Fi network (or laptop Mobile Hotspot).
2. On your mobile phone browser (Chrome or Safari), enter:
   http://192.168.31.94:5173
3. Dynamic Host Resolution: The frontend automatically detects the network hostname and routes all API calls to http://192.168.31.94:8000 without requiring any manual configuration.
4. If Windows Firewall blocks external mobile requests, run this 1-line PowerShell command as Admin:
   New-NetFirewallRule -DisplayName 'Janmitra Dev' -Direction Inbound -LocalPort 5173,8000 -Protocol TCP -Action Allow

PRE-CONFIGURED DEMO ACCOUNTS FOR EVALUATION

INSTANT TEST LOGIN

Role	Mobile Number	Password	Assigned Ward / Department	Permissions
Citizen	9900000021	citizen@123	Ward 1 (Dharampeth)	File issue, view GPS, rate repair
Citizen (Faizan)	9876543210	password123	Ward 4 (Mangalwari / North Nagpur)	Lead Developer Profile Account
Officer (PWD)	9900000001	officer@123	Roads & Infrastructure (4 Wards)	AI review, worker triage, SLA audit
Officer (SWM)	9900000002	officer@123	Solid Waste Management	Garbage triage, sanitary inspections
Field Worker	9900000005	worker@123	PWD Ward 1 (Road Maintenance)	Receive task push, upload proof photo
Super Admin	9900000000	admin@123	Nagpur Municipal Corporation	Citywide analytics & SLA oversight

OFFICIAL FINAL-YEAR CAPSTONE SIGN-OFF

PBCOE / RTMNU

Lead Developer Endorsement:

Faizan Rayeen (Final Year B.Tech)
Signature: _____

Project Guide Endorsement:

Department of Computer Science & Engg.
Signature: _____

Head of Department (CSE):

PBCOE Nagpur / RTMNU
Signature: _____